SCHÄFFER'S CONJECTURE FOR $k = 14$

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Abstract. We extend the results of Bennett-Györy-Pintér to prove Schäffer's conjecture for $k = 14$.

Contents

1. Introduction

Schäffer's conjecture [Sch56] concerns the following diophantine problem.

Conjecture 1.1. Let $S_k(x) = 1^k + 2^k + \cdots + x^k$. Then the only solutions to the equation

$$S_k(x) = y^n \tag{1}$$

in $x, y, k, n \in \mathbb{N}$ and $n \geq 2$ is the trivial solution $(x, y, k, n) = (1, 1, k, n)$ unless

$$(k, n) \in \{(1, 2), (3, 2), (3, 4), (5, 2)\},$$

in which case there are infinitely many solutions in $x, y \in \mathbb{N}$, or

$$(k, n, x, y) = (2, 2, 24, 70).$$

The conjecture has been proven in the following cases.

Theorem 1.2. *(Bennett-Györy-Pintér* [BGP04]*)* For $1 \leq k \leq 11$, equation (1) has only the trivial solution, unless $(k, n) \in \{(1, 2), (3, 2), (3, 4), (5, 2)\}$, or $k = 2$, in which case there is the additional solution $(n, x, y) = (2, 24, 70)$.

Date: October 26, 2025.

2020 *Mathematics Subject Classification.* 11D41; 11D61.

Bennett et al. showed this using a wide variety of methods. We extend this result with the following theorem.

Theorem 1.3. When $k = 14$, there are no solutions to (1).

A new ingredient in our proof is that we apply descent using all three factors $x, x+1, 2x-1$ present in the formula for $S_k(x)$ and we systematically apply the modular method, including the use of the multi-Frey technique, Kraus' method, and local methods modulo a power of a prime, and a result of Darmon-Merel [DM97]. A small number of cases then remain, but these are dealt with by prior results on ternary equations with coefficients [Ben01] [BBGP10].

If $n > 1$, then n is divisible by a prime p. Hence, to study (1), it suffices to consider $n = p$ prime.

2. Descent step

For even k, we may write

$$S_k(x) = \frac{1}{C_k} x(x+1)(2x+1)T_k(x) \tag{2}$$

where $C_k \in \mathbb{Z}$. Let $D = \gcd(x(x+1), T_k(x))$ and $d = \gcd(2x+1, T_k(x))$ where we consider the gcd over $\mathbb{Z}$ for a particular $x \in \mathbb{Z}$. For a fixed k, we will use the following result to get a finite list of possible values for D and d.

Lemma 2.1. Let $f = \alpha x + \beta, g = \gamma x^e + \cdots \in \mathbb{Z}[x]$, $\alpha, \gamma, \delta \neq 0$ be such that the absolute value of the resultant $|\mathrm{Res}(f, g, x)| = \delta$ where δ is squarefree. Then for all $x_0 \in \mathbb{Z}$, $\gcd(f(x_0), g(x_0)) \mid \delta$. Furthermore, suppose $\gcd(\alpha, \delta) = 1$ and $\delta \mid \alpha^{e-1} g'\left(-\frac{\beta}{\alpha}\right)$. Then if a prime $p \mid \gcd(f(x_0), g(x_0))$, then $p \parallel g(x_0)$.

Proof. Recall that there are polynomials $A(x)$ and $B(x)$ in $\mathbb{Z}[x]$ such that

$$A(x)f(x) + B(x)g(x) = \mathrm{Res}(f, g, x) = \alpha^e g\left(-\frac{\beta}{\alpha}\right).$$

Suppose $x_0 \in \mathbb{Z}$. We have that $\gcd(f(x_0), g(x_0)) \mid \delta$.

Suppose a prime $p \mid \gcd(f(x_0), g(x_0))$. Consider $\alpha^e g\left(\frac{x-\beta}{\alpha}\right)$ as a polynomial of degree e in $\mathbb{Z}[x]$ and write

$$\alpha^e g\left(\frac{x-\beta}{\alpha}\right) = \alpha^e g\left(-\frac{\beta}{\alpha}\right) + \alpha^{e-1} g'\left(-\frac{\beta}{\alpha}\right) x + x^2 p(x)$$

for some $p(x) \in \mathbb{Z}[x]$. Then for any $x \in \mathbb{Z}$, we have

$$\alpha^e g\left(\frac{px-\beta}{\alpha}\right) \equiv \alpha^e g\left(-\frac{\beta}{\alpha}\right) + \alpha^{e-1} g'\left(-\frac{\beta}{\alpha}\right)(px) \equiv \alpha^e g\left(-\frac{\beta}{\alpha}\right) = \mathrm{Res}(f, g, x) \pmod{p^2}$$

because $p \mid \delta$ and $\delta \mid \alpha^{e-1} g'\left(-\frac{\beta}{\alpha}\right)$. Therefore, $p^2 \nmid \alpha^e g\left(\frac{px-\beta}{\alpha}\right)$ because $\mathrm{Res}(f, g, x)$ is squarefree. We have proved that $p \| \alpha^e g\left(\frac{px-\beta}{\alpha}\right)$ for all $x \in \mathbb{Z}$. Since $p \mid f(x_0) = \alpha x_0 + \beta$, we write $x_0 = \frac{px-\beta}{\alpha}$ for some $x \in \mathbb{Z}$. So $p \| g(x_0)$ because $\gcd(\alpha, \delta) = 1$. This proves Lemma 2.1.

□

Corollary 2.2. For $2 \le k \le 14$ even, we have that

(i) D and d are both square-free,
(ii) if a prime $p \mid D$ or $p \mid d$, then $p \| T_k(x)$.

Proof. We apply Lemma 2.1 to each linear factor, $x, x+1, 2x+1$, where we note the three linear factors are pairwise coprime. We need to check the condition that $|\mathrm{Res}(f,g,x)| = \delta \mid \alpha^{\deg(T_k)} T_k'\left(-\frac{\beta}{\alpha}\right)$. For example, for $k = 14$, we have

$$T_{14}(x) = 3x^{12}+18x^{11}+24x^{10}-45x^9-81x^8+144x^7+182x^6-345x^5-217x^4+498x^3+44x^2-315x+105.$$

and $\mathrm{Res}(x, T_{14}(x), x) = 105 \mid T_{14}'(0) = -315$ and $\mathrm{Res}(x+1, T_{14}(x), x) = 105 \mid T_{14}'(-1) = 315$ and $\mathrm{Res}(2x+1, T_{14}(x), x) = 860055 = 3 \times 5 \times 7 \times 8191$ divides $2^{12} T_{14}'\left(-\frac{1}{2}\right) = 0$. □

The values for D and d are listed in Table 2. We will also need the following lemmas.

k	D divides	d divides	C_k
2	1	1	$2 \cdot 3$
4	1	7	$2 \cdot 3 \cdot 5$
6	1	31	$2 \cdot 3 \cdot 7$
8	3	$3 \cdot 127$	$2 \cdot 3^2 \cdot 5$
10	5	$5 \cdot 7 \cdot 73$	$2 \cdot 3 \cdot 11$
12	691	$23 \cdot 89 \cdot 691$	$2 \cdot 3 \cdot 5 \cdot 7 \cdot 13$
14	$3 \cdot 5 \cdot 7$	$3 \cdot 5 \cdot 7 \cdot 8191$	$2 \cdot 3^2 \cdot 5$

Table 1. Summary of constants

Lemma 2.3. Suppose
$$ab = dy^n$$
for $a, b, d, y, n \in \mathbb{N}$ where $\gcd(a,b) = 1$. Then there exist $d_1, d_2, y_1, y_2 \in \mathbb{N}$ such that
$$\begin{aligned} d &= d_1 d_2, \quad y = y_1 y_2 \\ \gcd(d_1, d_2) &= \gcd(y_1, y_2) = 1 \\ a/d_1 &= y_1^n, \quad b/d_2 = y_2^n. \end{aligned}$$

Proof. First note that d is a divisor of ab. Since $\gcd(a,b) = 1$, the divisors of ab are in bijection with (d_1, d_2), where d_1, d_2 are divisors of a, b, respectively. In this bijection, we have that $d_1 = \gcd(d,a)$, $d_2 = \gcd(d,b)$, and $d = d_1 d_2$. Now write

$$(a/d_1)(b/d_2) = y^n. \tag{3}$$

Since $\gcd(a/d_1, b/d_2) = 1$, we have that the valuation of each prime dividing the left hand side of (3) is divisible by n. Hence, by unique factorization in $\mathbb{Z}$, we obtain that $a/d_1 = y_1^n$ and $b/d_2 = y_2^n$ for some $y_1, y_2 \in \mathbb{N}$. □

Proposition 2.4. For even k, suppose
$$S_k(x) = \frac{1}{C_k} x(x+1)(2x+1)T_k(x) = y^n$$
has solution x, y, n in $\mathbb{N}$. Let $D = \gcd(x(x+1), T_k(x))$, $d = \gcd(2x+1, \frac{T_k(x)}{D})$ and $D_1 = \frac{D}{\gcd(D, C_k)}$. Assume

(i) D and d are both square-free,
(ii) if a prime $p \mid D$ or $p \mid d$, then $p \parallel T_k(x)$.

Then for some $\alpha, \beta, \alpha_0, \alpha_1, \beta_0, D_1, d_1 \in \mathbb{N}$ with $\alpha_0, \alpha_1, \beta_0, D_1, d_1$ coprime satisfying

$$\alpha\beta = \frac{C_k}{2\gcd(D, C_k)}, \quad \alpha_0\alpha_1 = 2\alpha, \quad \beta_0 \mid \frac{\beta}{\gcd(d,\beta)},$$

the system of equations

$$\alpha_1 w_1 - \alpha_0 w_0 - 1 = 0 \tag{4}$$

$$\beta_0 w_2 - 2\alpha_0 w_0 - 1 = 0 \tag{5}$$

$$2\alpha_1 w_1 - \beta_0 w_2 - 1 = 0 \tag{6}$$

is solvable in $u_0, u_1, v_0 \in \mathbb{N}$ where

$$w_0 = \begin{cases} (D_1)^{n-1} u_0^n & \text{if } (D_1)^{n-1} \mid x \\ u_0^n & \text{if } (D_1)^{n-1} \mid x+1 \end{cases}, \quad w_1 = \begin{cases} u_1^n & \text{if } (D_1)^{n-1} \mid x \\ (D_1)^{n-1} u_1^n & \text{if } (D_1)^{n-1} \mid x+1 \end{cases}, \quad w_2 = d_1^{n-1} v_0^n.$$

and $D_1 = \frac{D}{\gcd(D,C_k)}$ and $d_1 = \frac{d}{\gcd(d,\beta)}$. Further, $x = \alpha_0 w_0$, $x+1 = \alpha_1 w_1$, and $2x+1 = \beta_0 w_2$.

Proof. Note that since $D \mid T_k(x)$, we have $D \mid C_k y^n$. Next, define $D_1 := \frac{D}{\gcd(D,C_k)}$ so that $\gcd(D_1, C_k) = 1$. Since $D_1 \mid D, D \mid C_k y^n$ and $\gcd(D_1, C_k) = 1$, we have $D_1 \mid y^n$. Additionally, since D is squarefree, so is D_1, thus $D_1 \mid y$. Let $y_m := \frac{y}{D_1}$.

Multiplying (2) by $\frac{C_k}{D}$, we have

$$\begin{aligned} x(x+1)(2x+1)\frac{T_k(x)}{D} &= \frac{C_k y^n}{D} \\ &= \frac{C_k y^n}{\gcd(D, C_k) D_1} \\ &= \frac{C_k}{\gcd(D, C_k)} \frac{y^n}{D_1} \\ &= \frac{C_k}{\gcd(D, C_k)} (D_1)^{n-1} y_m^n. \end{aligned}$$

Since x, $x+1$, and $2x+1$ are pairwise coprime, and by Corollary 2.2,

$$\gcd\left(x(x+1), (2x+1)\frac{T_k(x)}{D}\right) = 1. \tag{7}$$

Next, we know $(D_1)^{n-1} \mid x(x+1)(2x+1)\frac{T_k(x)}{D}$. Since we have $D_1 \mid D \mid x(x+1)$, if $D_1 \mid (2x+1)\frac{T_k(x)}{D}$, then this would contradict (7). Hence $\gcd(D_1, (2x+1)\frac{T_k(x)}{D}) = 1$ so $(D_1)^{n-1} \mid x(x+1)$. Note that $2 \mid x(x+1)$. Then we have

$$\gcd\left(\frac{1}{2(D_1)^{n-1}} x(x+1), (2x+1)\frac{T_k(x)}{D}\right) = 1 \tag{8}$$

so by Lemma 2.3 we can write

$$y_m^n = u^n v^n, \tag{9}$$

where

$$u^n = \frac{1}{2(D_1)^{n-1}\alpha} x(x+1), \qquad v^n = \frac{1}{\beta}(2x+1)\frac{T_k(x)}{D} \tag{10}$$

where $\alpha\beta = \frac{C_k}{2\gcd(D, C_k)}$. Note that since $\gcd(x, x+1) = 1$, we must have either $\gcd(D_1, x) = 1$ or $\gcd(D_1, x+1) = 1$ and hence when $D_1 \neq 1$, we have (exclusively) $(D_1)^{n-1} \mid x$ or $(D_1)^{n-1} \mid x+1$. Again by Lemma 2.3, since $x, x+1$ are coprime, we can write

$$u^n = u_0^n u_1^n \tag{11}$$

where either

(a) $$u_0^n = \frac{1}{\alpha_0 (D_1)^{n-1}} x, \qquad u_1^n = \frac{1}{\alpha_1}(x+1)$$

when $D_1 \mid x$ or

(b) $$u_0^n = \frac{1}{\alpha_0} x, \qquad u_1^n = \frac{1}{\alpha_1 (D_1)^{n-1}}(x+1)$$

when $D_1 \mid x+1$, where $\alpha_0 \alpha_1 = 2\alpha$. Define

$$w_0 = \begin{cases} (D_1)^{n-1} u_0^n & \text{if } D_1 \mid x \\ u_0^n & \text{if } D_1 \mid x+1 \end{cases}, \qquad w_1 = \begin{cases} u_1^n & \text{if } D_1 \mid x \\ (D_1)^{n-1} u_1^n & \text{if } D_1 \mid x+1 \end{cases}$$

Rearranging and substituting the expressions for u_0^n and u_1^n, we get

$$\alpha_0 w_0 + 1 = \alpha_1 w_1. \tag{12}$$

It is clear that α_0, α_1, D_1 are coprime. Note that if $(D, d) \neq 1$, then there is a prime p such that $p \mid \gcd(x(x+1), 2x+1)$, which is a contradiction. Hence, $\gcd(d, D) = 1$.

Next, define $d_1 = \frac{d}{\gcd(d, \beta)}$. Since d is squarefree, so is d_1. We have $\gcd(d_1, \beta) = 1$ and $d_1 \mid \beta v^n$ so $d_1 \mid v^n$. Since d_1 is squarefree, $d_1 \mid v$. Let $v_d := \frac{v}{d_1}$. Then

$$\begin{aligned} (2x+1)\frac{T_k(x)}{dD} &= \frac{\beta v^n}{d} \\ &= \frac{\beta}{\gcd(d,\beta)} \frac{v^n}{d_1} \\ (2x+1)\frac{T_k(x)}{dD} &= \frac{\beta}{\gcd(d,\beta)} d_1^{n-1} v_d^n \end{aligned}$$

By Corollary 2.2, we have

$$\gcd\left(2x+1, \frac{T_k(x)}{dD}\right) = 1.$$

Since $d_1 \mid d$, $\gcd\left(d_1, \frac{T_k(x)}{dD}\right) = 1$. Hence $d_1^{n-1} \mid 2x+1$ so

$$\gcd\left(\frac{1}{d_1^{n-1}}(2x+1), \frac{T_k(x)}{dD}\right) = 1$$

So by Lemma 2.3, we can write $v_d^n = v_0^n v_1^n$ where

$$v_0^n := \frac{1}{d_1^{n-1}\beta_0}(2x+1), \qquad v_1^n := \frac{1}{\beta_1}\frac{T_k(x)}{dD} \tag{13}$$

and $\beta_0\beta_1 = \frac{\beta}{\gcd(d,\beta)}$. Define

$$w_2 = d_1^{n-1}v_0^n$$

Rearranging and substituting w_0 and w_1, we get

$$2\alpha_0 w_0 + 1 = \beta_0 w_2 \tag{14}$$

and

$$2\alpha_1 w_1 - 1 = \beta_0 w_2 \tag{15}$$

From each equation, we respectively see that $\alpha_0, D_1, \beta_0, d_1$ and $\alpha_1, D_1, \beta_0, d_1$ are pairwise coprime, proving the theorem. □

3. The modular method

Recall $n > 1$, then n is divisible by a prime p. Hence, to study (1), it suffices to consider $n = p$ prime. Using Proposition 2.4, solutions to (1) give rise to solutions to (4)–(6) which are of generalized Fermat type with one of the variables set to 1.

We now briefly summarize the modular method as it is applied to resolve generalized Fermat type equations of the form

$$Ax^p + By^p + Cz^p = 0, \tag{16}$$

where A, B, C are non-zero pairwise coprime integers which are not pth powers and such that $p \nmid ABC$. Our treatment follows [HK02].

Up to permutation of the terms and negation in (16), we can assume that By^p is even and $Ax^p \equiv -1 \pmod 4$.

To a primitive non-trivial solution (a, b, c) of (16) we attach the Frey elliptic curve

$$Y^2 = X(X - Aa^p)(X + Bb^p) \tag{17}$$

Consider the Galois representation

$$\overline{\rho}_{E,p} : G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{F}_p)$$

attached to the elliptic curve E over $\mathbb{Q}$, which is obtained by letting $G_{\mathbb{Q}} \coloneqq \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ act linearly on $E[p]$. As the E is a Frey elliptic curve over $\mathbb{Q}$, we have by modularity and level lowering that

$$\overline{\rho}_{E,p} \simeq \overline{\rho}_{f,\mathfrak{p}} \tag{18}$$

where f is a classical newform form of weight 2, level N equal to the Serre level of $\overline{\rho}_{E,p}$, and trivial character and $\mathfrak{p}$ is a prime of the coefficient field K of f above p.

By Kraus-Halberstadt [HK02], we can compute the Serre level N of $\overline{\rho}_{E,p}$. Let $R = ABC$. Then

$$N = \begin{cases} 2\mathrm{rad}'(R) & v_2(R) = 0 \text{ or } v_2(R) \geq 5 \\ 2\mathrm{rad}'(R) & 1 \leq v_2(R) \leq 3, xyz \text{ even} \\ \mathrm{rad}'(R) & v_2(R_i) = 4 \\ 32\mathrm{rad}'(R) & v_2(R) = 1, xyz \text{ odd} \\ 8\mathrm{rad}'(R) & v_2(R) \in \{2,3\}, xyz \text{ odd} \end{cases}. \tag{19}$$

The conductor N_E of $E/\mathbb{Q}$ is square-free except powers of 2. Hence, the Frey elliptic curve has either good or multiplicative bad reduction modulo primes $\ell \neq 2$.

Computing the finite set of newforms f of weight 2, level N, and trivial character we now consider the isomorphism in (18) for each possible f. The idea behind the modular method is to show this is not possible for each f, i.e. eliminate every newform f for p sufficiently large.

Let $\ell \nmid 2Np$ be a prime so that $\overline{\rho}_{E,p}$ is unramified at ℓ. Let $\mathrm{Frob}_\ell \in G_{\mathbb{Q}} \coloneqq \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ be a Frobenius element at ℓ. We may thus consider the well defined trace value

$$a_\ell(f) = \mathrm{tr}\, \rho_{f,\mathfrak{p}}(\mathrm{Frob}_\ell) \in \mathcal{O}_K, \tag{20}$$

where $\rho_{f,\mathfrak{p}}$ is the $\mathfrak{p}$-adic representation attached to f.

If $\ell \nmid N_E$, where N_E is the conductor of $E/\mathbb{Q}$, then E has good reduction at ℓ and we may consider

$$a_\ell(E) = \mathrm{tr}\, \rho_{E,p}(\mathrm{Frob}_\ell) \in \mathbb{Z}, \tag{21}$$

where $\rho_{E,p}$ is the p-adic representation attached to E.

The isomorphism (18) implies that

$$a_\ell(E) \equiv a_\ell(f) \pmod{\mathfrak{p}}, \tag{22}$$

and hence

$$\mathrm{Norm}_{K/\mathbb{Q}}(a_\ell(E) - a_\ell(f)) \equiv 0 \pmod{p} \tag{23}$$

where $\mathrm{Norm}_{K/\mathbb{Q}}(\cdot)$ is the norm from K to $\mathbb{Q}$.

If $\ell \mid N_E$, then E has multiplicative bad reduction at ℓ. It can be shown that $\mathrm{WD}_\ell(\{\rho_{E,p}\})$ is a special representation with character being an unramified character of order dividing 2. It follows that

$$\pm(\ell+1) \equiv a_\ell(f) \pmod{\mathfrak{p}}, \tag{24}$$

and hence

$$\text{Norm}_{K/\mathbb{Q}}((\ell+1)^2 - a_\ell(f)^2) \equiv 0 \pmod{p}. \tag{25}$$

Proposition 3.1. Let (x,y,p) be a solution to

$$Ax^p - By^p = 1 \tag{26}$$

where A,B,C are pairwise coprime. WLOG, we may assume $By^p \equiv 0 \pmod 2$. Let $\mathcal{N}$ be the set of possible levels given by (19), ℓ a prime, and W_ℓ be the set of solutions to (26) mod ℓ. For $w = (w_0, w_1, n) \in W_\ell$, let $E(w) : y^2 = X(X+1)(X - Bw_1^n)$ and f a newform of weight 2 and level N. Define

$$B_\ell(f,w) := \begin{cases} \text{Norm}_{K/\mathbb{Q}}(a_\ell(E(w)) - a_\ell(f)) & E(w) \text{ has good reduction at } \ell, \\ \text{Norm}_{K/\mathbb{Q}}((\ell+1)^2 - a_\ell(f))^2) & E(w) \text{ has bad reduction at } \ell. \end{cases}$$

Then,

$$p \in \bigcup_{N\in\mathcal{N}} \bigcup_{\substack{f \text{ modular} \\ \text{level } N}} \bigcap_{\substack{\ell\nmid N \\ \text{prime}}}^{L} \bigcup_{w\in W_\ell} \{p \text{ prime} : p \mid B_\ell(f,w)\} \cup \{\ell\}$$

for any bound $L \in \mathbb{N}$.

Proof. If (x,y,p) is a solution to (26) then by the above discussion, there is a newform form f of weight 2 with level $N \in \mathcal{N}$ such that $\overline{\rho}_{E,\ell} \simeq \overline{\rho}_{f,\ell}$, where $E : y^2 = X(X+1)(X - By^p)$. Since (x,y,p) is a solution to (26), $\exists w = (w_0, w_1) \in W_\ell$ where $w_0 \equiv x^p \pmod{\ell}$ and $w_1 \equiv y^p \pmod{\ell}$. By (23) and (25), we have thus have $p \mid B_\ell(f,w)$ or $p = \ell$. In particular,

$$p \in \{p \text{ prime} : p \mid B_\ell(f,w)\} \cup \{\ell\}$$

and thus

$$p \in \bigcup_{N\in\mathcal{N}} \bigcup_{\substack{f \text{ modular} \\ \text{level } N}} \bigcap_{\substack{\ell\nmid N \\ \text{prime}}} \bigcup_{w\in W_\ell} \{p \text{ prime} : p \mid B_\ell(f,w)\} \cup \{\ell\}$$

$\square$

In particular, if for all possible levels $N \in \mathcal{N}$ and all weight 2 newforms f of level N, there exists $\ell \nmid N$ prime such that $\forall w \in W_\ell$, $\{p \text{ prime} : p \mid B_\ell(f,w)\}$ is finite, then we have that p is one of only finitely many possibilities.

The Frey elliptic curve E_i associated to each generalized Fermat type equation (4)–(6) is given by:

$$E_0 : y^2 = \begin{cases} x(x+1)(x - \alpha_0 w_0) & \alpha_0 \equiv 0 \pmod 2 \\ x(x+1)(x + \alpha_1 w_1) & \alpha_1 \equiv 0 \pmod 2 \end{cases}, \tag{27}$$

$$E_1 : y^2 = x(x+1)(x - 2\alpha_0 w_0), \tag{28}$$

$$E_2 : y^2 = x(x+1)(x + 2\alpha_1 w_1). \tag{29}$$

The quantities R_i are given by:

$$R_0 = \alpha_0\alpha_1 D_1, \quad R_1 = \begin{cases} 2\alpha_0 D_1 \beta_0 d_1 & \text{if } D_1 \mid x \\ 2\alpha_0 \beta_0 d_1 & \text{if } D_1 \mid x+1 \end{cases}, \quad R_2 = \begin{cases} 2\alpha_1 \beta_0 d_1 & \text{if } D_1 \mid x \\ 2\alpha_1 D_1 \beta_0 d_1 & \text{if } D_1 \mid x+1 \end{cases} \tag{30}$$

from which we can compute the Serre levels N_i of $\overline{\rho}_{E_i,p}$ by (19).

For a newform f_i of level N_i, let $K_i = \mathbb{Q}(a_n(f_i) : n \in \mathbb{N})$. Recall from (18) and above, there exists a prime ideal $\mathfrak{p}_i$ of K_i over p such that for all primes $\ell \neq p$ and $\ell \nmid N_i$,

(1) if E_i is non-singular mod ℓ, then $a_\ell(E_i) \equiv a_\ell(f_i) \pmod{\mathfrak{p}_i}$,
(2) if E_i is singular mod ℓ, then $\pm(\ell+1) \equiv a_\ell(f_i) \pmod{\mathfrak{p}_i}$.

In particular, define

$$B_\ell^{(i)}(f) := \begin{cases} \mathrm{Norm}_{K_i/\mathbb{Q}}(a_\ell(E_i) - a_\ell(f)) & E_i \text{ non-singular mod } \ell, \\ \mathrm{Norm}_{K_i/\mathbb{Q}}((\ell+1)^2 - a_\ell(f))^2) & E_i \text{ singular mod } \ell. \end{cases}$$

Thus any solution to (4)–(6) must have exponent $p \mid B_i(f)$ or $p = \ell$. Note that since this holds for all primes $\ell \neq p$, we can consider $B_i(f)$ for various ℓ and take the intersection of primes which appear in $B_i(f)$ to restrict the exponent p: we have a finite bound on the exponent p provided $B_i(f) \not\equiv 0$.

Corollary 3.2. Let (x, y, p) be a non-trivial solution to (1). Let $\mathcal{N}_i$ be the set of possible levels to the elliptic curves in (27) given by Kraus, $\mathcal{N} := \mathcal{N}_0 \cup \mathcal{N}_1 \cup \mathcal{N}_2$, ℓ a prime, and W_ℓ be the set of solutions of (4),(5),(6) respectively given by (2.4) mod $\ell \nmid \prod_{N_i \in \mathcal{N}}$. For $w = (w_0, w_1, w_2, p) \in W_\ell$, let

$$\begin{aligned} E_0(w) &: y^2 = \begin{cases} x(x+1)(x - \alpha_0 w_0) & \alpha_0 \equiv 0 \pmod 2 \\ x(x+1)(x + \alpha_1 w_1) & \alpha_1 \equiv 0 \pmod 2 \end{cases}, \\ E_1(w) &: y^2 = x(x+1)(x - 2\alpha_0 w_0), \\ E_2(w) &: y^2 = x(x+1)(x + 2\alpha_1 w_1). \end{aligned}$$

and let f_i a newform of weight 2 and level $N_i \in \mathcal{N}$. Define

$$B_\ell^{(i)}(f_i, w) := \begin{cases} \mathrm{Norm}_{K_i/\mathbb{Q}}(a_\ell(E_i(w)) - a_\ell(f_i)) & E_i(w) \text{ has good reduction at } \ell, \\ \mathrm{Norm}_{K_i/\mathbb{Q}}((\ell+1)^2 - a_\ell(f_i))^2) & E_i(w) \text{ has bad reduction at } \ell. \end{cases}$$

and let

$$A_\ell^{(i)}(w) = \{p \text{ prime} : p \mid B_\ell^{(i)}(f_i, w) \text{ for some newform } f_i \text{ of level } N_i \in \mathcal{N}_i\}$$

for $i = 1, 2$. Then

$$p \in \bigcup_{N_0 \in \mathcal{N}_0} \bigcup_{\substack{f_0 \text{ modular} \\ \text{level } N_0}} \bigcap_{\substack{\ell \nmid N \\ N \in \mathcal{N}}} \bigcup_{w \in W_\ell} \Big(\{p \text{ prime} : p \mid B_\ell^{(0)}(f_0, w)\} \cap A_1(w) \cap A_2(w) \Big) \cup \{\ell\}$$

Proof. For a non-trivial solution (x, y, p) to (1), by (2.4) we get a system

$$\begin{aligned} \alpha_1 w_1 - \alpha_0 w_0 &= 1 \\ \beta_0 w_2 - 2\alpha_0 w_0 &= 1 \\ 2\alpha_1 w_1 - \beta_0 w_2 &= 1 \end{aligned}$$

where

$$w_0 = \begin{cases} (D_1)^{n-1} u_0^n & \text{if } (D_1)^{n-1} \mid x \\ u_0^n & \text{if } (D_1)^{n-1} \mid x+1 \end{cases}, \qquad w_1 = \begin{cases} u_1^n & \text{if } (D_1)^{n-1} \mid x \\ (D_1)^{n-1} u_1^n & \text{if } (D_1)^{n-1} \mid x+1 \end{cases}, \qquad w_2 = d_1^{n-1} v_0^n.$$

where $\alpha_i, \beta_0, D_1, d_1$ are pairwise coprime. Applying (3.1) to (4), (5), and (6), we have that

$$p \in \bigcup_{N_i \in \mathcal{N}_i} \bigcup_{\substack{f_i \text{ modular} \\ \text{level } N_i}} \bigcap_{\substack{\ell \nmid N_i \\ \text{prime}}}^{L} \bigcup_{w \in W_\ell^{(i)}} \{p \text{ prime} : p \mid B_\ell^{(i)}(f_i, w)\} \cup \{\ell\}$$

for $i = 0, 1, 2$ respectively, where $W_\ell^{(i)}$ are the solutions to (4), (5), and (6). Since the solutions (x, y, p) must satisfy all three equations simultaneously, for a given prime ℓ, any solution w to the system mod ℓ must be in $W_\ell = W_\ell^{(0)} \cap W_\ell^{(1)} \cap W_\ell^{(2)}$. Thus,

$$p \in \bigcup_{N_i \in \mathcal{N}_i} \bigcup_{\substack{f_i \text{ modular} \\ \text{level } N_i}} \bigcap_{\substack{\ell \nmid N_i \\ \text{prime}}}^{L} \bigcup_{w \in W_\ell} \{p \text{ prime} : p \mid B_\ell^{(i)}(f_i, w)\} \cup \{\ell\}$$

Note

$$\begin{aligned}
&\bigcup_{N_i \in \mathcal{N}_i} \bigcup_{\substack{f_i \text{ modular} \\ \text{level } N_i}} \bigcap_{\substack{\ell \nmid N_i \\ \text{prime}}}^{L} \bigcup_{w \in W_\ell} \{p \text{ prime} : p \mid B_\ell^{(i)}(f_i, w)\} \cup \{\ell\} \\
&\subseteq \bigcap_{\ell \nmid \prod_{N_i \in \mathcal{N}_i} N}^{L} \bigcup_{w \in W_\ell} \bigcup_{N_i \in \mathcal{N}} \bigcup_{\substack{f_i \text{ modular} \\ \text{level } N_i}} \{p \text{ prime} : p \mid B_\ell^{(i)}(f_i, w)\} \cup \{\ell\} \\
&= \bigcap_{\ell \nmid \prod_{N_i \in \mathcal{N}_i} N}^{L} \bigcup_{w \in W_\ell} A_\ell^{(i)}(w) \cup \{\ell\} \\
&\subseteq \bigcap_{\ell \nmid \prod_{N \in \mathcal{N}} N}^{L} \bigcup_{w \in W_\ell} A_\ell^{(i)}(w) \cup \{\ell\}
\end{aligned}$$

Hence,

$$\begin{aligned}
p \in &\bigcap_{i=0}^{2} \bigcup_{N_i \in \mathcal{N}_i} \bigcup_{\substack{f_i \text{ modular} \\ \text{level } N_i}} \bigcap_{\substack{\ell \nmid N_i \\ \text{prime}}}^{L} \bigcup_{w \in W_\ell} \{p \text{ prime} : p \mid B_\ell^{(i)}(f_i, w)\} \cup \{\ell\} \\
&\subseteq \left(\bigcup_{N_0 \in \mathcal{N}_0} \bigcup_{\substack{f_0 \text{ modular} \\ \text{level } N_0}} \bigcap_{\substack{\ell \nmid N \\ N \in \mathcal{N}}}^{L} \bigcup_{w \in W_\ell} \{p \text{ prime} : p \mid B_\ell^{(0)}(f_0, w)\} \cup \{\ell\} \right) \cap \left(\bigcap_{i=0}^{1} \bigcap_{\substack{\ell \nmid N \\ N \in \mathcal{N}}}^{L} \bigcup_{w \in W_\ell} A_\ell^{(i)}(w) \cup \{\ell\} \right) \\
&= \bigcup_{N_0 \in \mathcal{N}_0} \bigcup_{\substack{f_0 \text{ modular} \\ \text{level } N_0}} \bigcap_{\substack{\ell \nmid N \\ N \in \mathcal{N}}}^{L} \bigcup_{w \in W_\ell} \Big(\{p \text{ prime} : p \mid B_\ell^{(0)}(f_0, w)\} \cap A_1(w) \cap A_2(w) \Big) \cup \{\ell\}
\end{aligned}$$

□

In particular, if for every possible system of equations given by (2.4), we can show that the set given by (3.2) is empty, then there are no solutions to (1).

Theorem 3.3. *(Darmon-Merel [DM97])* The only non-zero primitive solutions (a, b, c) to the equation

(31) $$x^n + y^n = 2z^n$$

satisfies $abc = \pm 1$.

Corollary 3.4. Suppose (x, y, n) is a solution to (1). By 2.4, there exist $\alpha_0, \alpha_1, \beta_0, D_1, d_1 \in \mathbb{N}$ pairwise coprime satisfying (4), (5), and (6). None of the following are true.

(1) $\alpha_0\alpha_1 = 2$ and $D_1 = 1$,
(2) $\alpha_0\beta_0 d_1 D_1 = 1$, or
(3) $\alpha_0\beta_0 d_1 D_1 = 1$

Proof. If $\alpha_0\alpha_1 = 2$ and $D_1 = 1$, then (4) is

$$2u_0^n + 1 = u_1^n \quad \text{or} \quad u_0^n + 1 = 2u_1^n$$

In either case, by Theorem 3.3, the only solution satisfies $|u_0u_1| = 1$. If $\alpha_0 = 1$, then $u_0 = u_1 = 1$ so $x = \alpha_0 u_0^n = 1$, which is the trivial solution. If $\alpha_1 = 1$, then $u_0 = u_1 = -1$ so $x = \alpha_0 u_0^n = -2 \notin \mathbb{N}$. If $\alpha_0\beta_0 d_1 D_1 = 1$, then (5) is

$$2u_0^n + 1 = v_0^n$$

and if $\alpha_1\beta_0 d_1 D_1 = 1$, then (6) is

$$2u_1^n - 1 = v_0^n$$

so we get $u_0 = -1$ and $u_1 = 1$ respectively, which as before leads to a contradiction. □

4. Local constraints

To eliminate a specific prime p from being a solution to a generalized Fermat equation, we try the following method of Kraus [Kra98].

Corollary 4.1. Let (x, y, p) be a solution to (1). Let $\ell \equiv 1 \pmod p$ be a prime. Let W_ℓ be the set of p-th power solutions (w_0, w_1, w_2, p) to the system given by (2.4) mod ℓ and $\mathcal{N}$ be the set of possible levels given the equations (4), (5), and (6). Then

$$p \in \bigcap_{i=0}^{2} \bigcup_{N_i \in \mathcal{N}_i} \bigcup_{\substack{f_i \text{ modular} \\ \text{level } N_i}} \bigcap_{\substack{\ell \nmid N_i \\ \text{prime}}}^{L} \bigcup_{w \in W_\ell} \{p \text{ prime} : p \mid B_\ell^{(i)}(f_i, w)\}$$

where $B_\ell^{(i)}(w)$ is given in (3.2).

This follows immediately from (3.1). Being a pth power mod ℓ, where $\ell \equiv 1 \pmod p$, is a very restrictive constraint. For example, if $\ell = 2p+1$, then the only pth powers modulo ℓ are ± 1.

Lemma 4.2. If (x, y, p) is a solution to (1), then there exists a solution (w_0, w_1, w_2, p) to the system from (2.4) modulo p^2.

In particular, if we can show that there are no solutions (w_0, w_1, w_2, p) modulo p^2, then we can eliminate p from being a possible exponent.

Let F be the map that takes a finite set of primes S and returns the set of primes not eliminated by either (4.1) or (4.2), i.e., If K is the set given by (4.1) and W_{p^2} is the set of solutions to the system from (2.4) modulo p^2, then

$$F(S) = \left\{p \in S : p \in K \text{ and } W_{p^2} \neq \varnothing\right\}$$

Corollary 4.3. Considering the same setup as (3.2), we have that

$$p \in \bigcup_{N_0 \in \mathcal{N}_0} \bigcup_{\substack{f_0 \text{ modular} \\ \text{level } N_0}} \bigcap_{\substack{\ell \nmid N \\ N \in \mathcal{N}}}^{L} F\left(\bigcup_{w \in W_\ell} \left(\{p \text{ prime} : p \mid B_\ell^{(0)}(f_0, w)\} \cap A_1(w) \cap A_2(w)\right) \cup \{\ell\}\right)$$

Proof. If (x, y, p) is a solution to (1), then by (3.2), we have that

$$p \in \bigcup_{N_0 \in \mathcal{N}_0} \bigcup_{\substack{f_0 \text{ modular} \\ \text{level } N_0}} \bigcap_{\substack{\ell \nmid N \\ N \in \mathcal{N}}}^{L} \bigcup_{w \in W_\ell} \left(\{p \text{ prime} : p \mid B_\ell^{(0)}(f_0, w)\} \cap A_1(w) \cap A_2(w)\right) \cup \{\ell\} \tag{32}$$

and thus we must have

$$p \in F\left(\bigcup_{N_0 \in \mathcal{N}_0} \bigcup_{\substack{f_0 \text{ modular} \\ \text{level } N_0}} \bigcap_{\substack{\ell \nmid N \\ N \in \mathcal{N}}}^{L} \bigcup_{w \in W_\ell} \left(\{p \text{ prime} : p \mid B_\ell^{(0)}(f_0, w)\} \cap A_1(w) \cap A_2(w)\right) \cup \{\ell\}\right)$$

Note that F is independent of the choice of level N_0, modular form f_0 and prime ℓ. That is, if p is in (32) and either $p \notin K$ or $W_{p^2} = \varnothing$, then this holds independent of N_0, f_0 and ℓ. Thus

$$\begin{aligned} &F\left(\bigcup_{N_0 \in \mathcal{N}_0} \bigcup_{\substack{f_0 \text{ modular} \\ \text{level } N_0}} \bigcap_{\substack{\ell \nmid N \\ N \in \mathcal{N}}}^{L} \bigcup_{w \in W_\ell} \left(\{p \text{ prime} : p \mid B_\ell^{(0)}(f_0, w)\} \cap A_1(w) \cap A_2(w)\right) \cup \{\ell\}\right) \\ &= \bigcup_{N_0 \in \mathcal{N}_0} \bigcup_{\substack{f_0 \text{ modular} \\ \text{level } N_0}} \bigcap_{\substack{\ell \nmid N \\ N \in \mathcal{N}}}^{L} F\left(\bigcup_{w \in W_\ell} \left(\{p \text{ prime} : p \mid B_\ell^{(0)}(f_0, w)\} \cap A_1(w) \cap A_2(w)\right) \cup \{\ell\}\right) \end{aligned}$$

□

5. Linear forms in logarithms

We will make use of two powerful theorems to deal with any cases not handled by the modular and local methods.

Theorem 5.1. [Ben01] If $A, B, n \in \mathbb{Z}$, $AB \neq 0$, $n \geq 3$, then

$$|Ax^n - By^n| = 1$$

has at most one solution in positive integers (x, y).

In particular, if $A = B + 1$, then the only solution is $(x, y) = (1, 1)$.

Theorem 5.2. [BBGP10] If $1 < B \leq 400$, then all integer solutions (x, y, n) of

$$|x^n - By^n| = 1$$

with $|xy| > 1, n \geq 3$ and with $(B, n) \notin \{(235, 23), (282, 23), (295, 29), (329, 23), (354, 29)\}$ are given by

$$\begin{aligned} n = 3,\ (B, x, y) = &(7, \pm(2,1)), (9 \pm (2,1)), (17, \pm(18,7)), (19, \pm(8,3)), \\ &(20 \pm (19,7)), (26, \pm(3,1), (63, \pm(4,1)), (91, \pm(9,2)), (124, \pm(5,1)), \\ &(126, \pm(5,1)), (182, \pm(17,3)), (215, \pm(6,1)), (217, \pm(6,1)), \\ &(254, \pm(19,3)), (342, \pm(7,1)), (344, \pm(7,1)) \end{aligned}$$

$n = 4,\ (B, x, y) = (5, \pm 3, \pm 2), (15, \pm 2, \pm 1), (17, \pm 2, \pm 1), (39, \pm 5, \pm 2),$
$(80, \pm 3, \pm 1), (150, \pm 7, \pm 2), (255, \pm 4, \pm 1).$
$n = 5,\ (B, x, y) = (31, \pm(2, 1)), (242, \pm(3, 1)), (244, \pm(3, 1))$
$n = 6,\ (B, x, y) = (63, \pm 2, \pm 1)$
$n = 7,\ (B, x, y) = (127, \pm(2, 1)), (129, \pm(2, 1)),$
$n = 8,\ (B, x, y) = (255, \pm 2, \pm 1)$

6. The even cases $k = 2, 4, 6, 8, 14$ and n prime

By (2.4), for a fixed even k, if we can show that for all possible combinations of $\alpha_0, \alpha_1, \beta_0, D_1, d_1$ have no solutions, then there are no solutions to (1). Fix $k \in \{4, 6, 8, 14\}$ and $\alpha_0, \alpha_1, \beta_0, d_1, D_1$. If this case satisfies (3.4), then we move on. Otherwise, we apply either (4.3) to the whole system, or (3.1) to just one of the equations when the possible levels to some of the equations are computationally too large, with the bound $L = 1000$. We find that for each case, the only possible exponent is $p = 2$, except for the cases listed in the following table.

k	n	Case
$k = 4, 8, 14$	$n \in \mathbb{N}$	$\alpha_0 = 2, \alpha_1 = 3, \beta_0 = 5, D_1 = 1 = d_1$
		$\alpha_0 = 3, \alpha_1 = 2, \beta_0 = 5, D_1 = 1 = d_1$
$k = 8, 14$	$n \in \mathbb{N}$	$\alpha_0 = 2, \alpha_1 = 9, \beta_0 = 5, D_1 = 1 = d_1$
		$\alpha_0 = 9, \alpha_1 = 2, \beta_0 = 5, D_1 = 1 = d_1$
$k = 14$	$n = 3$	$\alpha_0 = 1, \alpha_1 = 18, \beta_0 = 1, D_1 = 7, d_1 = 1$
		$\alpha_0 = 18, \alpha_1 = 1, \beta_0 = 1, D_1 = 7, d_1 = 1$
		$\alpha_0 = 1, \alpha_1 = 90, \beta_0 = 1, D_1 = 7, d_1 = 1$
		$\alpha_0 = 90, \alpha_1 = 1, \beta_0 = 1, D_1 = 7, d_1 = 1$
		$\alpha_0 = 1, \alpha_1 = 6, \beta_0 = 1, D_1 = 7, d_1 = 1$
		$\alpha_0 = 6, \alpha_1 = 1, \beta_0 = 1, D_1 = 7, d_1 = 1$
		$\alpha_0 = 1, \alpha_1 = 10, \beta_0 = 1, D_1 = 7, d_1 = 1$
		$\alpha_0 = 10, \alpha_1 = 1, \beta_0 = 1, D_1 = 7, d_1 = 1$
		$\alpha_0 = 1, \alpha_1 = 30, \beta_0 = 1, D_1 = 7, d_1 = 1$
		$\alpha_0 = 30, \alpha_1 = 1, \beta_0 = 1, D_1 = 7, d_1 = 1$
		$\alpha_0 = 1, \alpha_1 = 18, \beta_0 = 1, D_1 = 7, d_1 = 1$
		$\alpha_0 = 18, \alpha_1 = 1, \beta_0 = 1, D_1 = 7, d_1 = 1$
		$\alpha_0 = 1, \alpha_1 = 6, \beta_0 = 1, D_1 = 7, d_1 = 1$
		$\alpha_0 = 6, \alpha_1 = 1, \beta_0 = 1, D_1 = 7, d_1 = 1$
		$\alpha_0 = 1, \alpha_1 = 10, \beta_0 = 1, D_1 = 7, d_1 = 1$
		$\alpha_0 = 10, \alpha_1 = 1, \beta_0 = 1, D_1 = 7, d_1 = 1$
		$\alpha_0 = 1, \alpha_1 = 18, \beta_0 = 1, D_1 = 7, d_1 = 1$
		$\alpha_0 = 18, \alpha_1 = 1, \beta_0 = 1, D_1 = 7, d_1 = 1$
		$\alpha_0 = 1, \alpha_1 = 6, \beta_0 = 1, D_1 = 7, d_1 = 1$
		$\alpha_0 = 6, \alpha_1 = 1, \beta_0 = 1, D_1 = 7, d_1 = 1$
		$\alpha_0 = 1, \alpha_1 = 30, \beta_0 = 1, D_1 = 1, d_1 = 8191$
		$\alpha_0 = 3, \alpha_1 = 10, \beta_0 = 1, D_1 = 1, d_1 = 8191$
		$\alpha_0 = 10, \alpha_1 = 3, \beta_0 = 1, D_1 = 1, d_1 = 8191$
		$\alpha_0 = 30, \alpha_1 = 1, \beta_0 = 1, D_1 = 1, d_1 = 8191$
		$\alpha_0 = 1, \alpha_1 = 18, \beta_0 = 1, D_1 = 7, d_1 = 8191$
		$\alpha_0 = 18, \alpha_1 = 1, \beta_0 = 1, D_1 = 7, d_1 = 8191$
		$\alpha_0 = 9, \alpha_1 = 10, \beta_0 = 1, D_1 = 7, d_1 = 8191$
		$\alpha_0 = 10, \alpha_1 = 9, \beta_0 = 1, D_1 = 7, d_1 = 8191$
		$\alpha_0 = 1, \alpha_1 = 6, \beta_0 = 1, D_1 = 7, d_1 = 8191$
		$\alpha_0 = 6, \alpha_1 = 1, \beta_0 = 1, D_1 = 7, d_1 = 8191$
		$\alpha_0 = 1, \alpha_1 = 10, \beta_0 = 1, D_1 = 7, d_1 = 8191$
		$\alpha_0 = 10, \alpha_1 = 1, \beta_0 = 1, D_1 = 7, d_1 = 8191$
		$\alpha_0 = 1, \alpha_1 = 30, \beta_0 = 1, D_1 = 7, d_1 = 8191$
		$\alpha_0 = 30, \alpha_1 = 1, \beta_0 = 1, D_1 = 7, d_1 = 8191$

In the first four exceptional cases, we could not eliminate any primes using the bound $L = 1000$, so we instead appeal to (5.1).

In the first and third exceptional cases, we have $\alpha_0 = 2, \beta_0 = 5$, so (5) leads to

$$4u_0^n + 1 = 5v_0^n$$

By (5.1), the only solution is $(u_0, v_0) = (1,1)$. Then $x = \alpha_0 u_0^n = 2$. Also, $S_4(2) = 1 + 2^4 = 17$, $S_8(2) = 1 + 2^8 = 257$, and $S_{14}(2) = 16385 = 5 \cdot 29 \cdot 113$, all of which are squarefree so we must have $n = 1$. But we assumed $n = p$ is prime, so there are no solutions in these cases. In the second and fourth exceptional cases, we have $\alpha_1 = 2, \beta_0 = 5, d_1 = 1$, so (6) leads to

$$4u_1^n - 1 = 5v_0^n$$

and by Bennett [Ben01], the only solution is $(u_0, v_0) = (-1,-1)$, so $x = \alpha_1 u_1^n - 1 < 0$, which is a contradiction.

In the remaining exceptional cases, all primes except for $p \in \{2,3\}$ were eliminated. In the cases with $\beta_0 = 1 = d_1$, in the case where $D_1 \mid x$, (16) leads to

$$2\alpha_1 u_1^n - v_0^n = 1$$

and in the case where $D_1 \mid x+1$, (15) leads to

$$v_0^n - 2\alpha_0 u_0^n = 1$$

In either case [BBGP10] shows that there are no solutions when $n = 3$ and $2\alpha_i \mid 90$.

Lemma 6.1. The only solutions in positive integers x, y to $Ax^3 - By^3 = 1$ when $\gcd(A,B) = 1$, $A \neq B+1$, $AB \neq 0$ and $\max\{A,B\} \leq 1000$ are given by table 1 in the appendix. Additionally, there are no solutions when $A = 1, B = 7^2 \cdot 30$ or $A = 7^2 \cdot 30, B = 1$.

$$\begin{aligned}(A,B,x,y) = {} & (1,7,2,1), (1,17,18,7), (1,26,3,1), (1,37,10,3), (1,63,4,1), (1,91,9,2), (1,124,5,1),\\ & (1,215,6,1), (1,254,19,3), (1,342,7,1), (1,511,8,1), (1,614,17,2), (1,635,361,42),\\ & (1,728,9,1), (1,813,28,3), (1,999,10,1), (2,15,2,1), (2,53,3,1), (2,127,4,1), (2,249,5,1),\\ & (2,431,6,1), (2,685,7,1), (3,10,3,2), (3,23,2,1), (3,80,3,1), (3,191,4,1), (3,292,23,5),\\ & (3,374,5,1), (3,499,11,2), (3,647,6,1), (4,31,2,1), (4,107,3,1), (4,255,4,1), (4,499,5,1),\\ & (4,863,6,1), (5,39,2,1), (5,78,5,2), (5,134,3,1), (5,319,4,1), (5,624,5,1), (6,47,2,1),\\ & (6,161,3,1), (6,383,4,1), (6,749,5,1), (7,55,2,1), (7,188,3,1), (7,300,7,2), (7,447,4,1),\\ & (7,811,39,8), (7,874,5,1), (8,17,9,7), (8,37,5,3), (8,63,2,1), (8,215,3,1), (8,511,4,1),\\ & (8,813,14,3), (8,999,5,1), (9,1,1,2), (9,71,2,1), (9,242,3,1), (9,575,4,1), (9,820,9,2),\\ & (10,79,2,1), (10,127,7,3), (10,269,3,1), (10,639,4,1), (11,19,6,5), (11,37,3,2),\\ & (11,87,2,1), (11,296,3,1), (11,703,4,1), (12,95,2,1), (12,323,3,1), (12,767,4,1),\\ & (13,5,8,11), (13,103,2,1), (13,203,5,2), (13,350,3,1), (13,746,27,7), (13,831,4,1),\\ & (14,111,2,1), (14,377,3,1), (14,895,4,1), (15,119,2,1), (15,404,3,1), (15,643,7,2),\\ & (15,791,15,4), (15,959,4,1), (16,127,2,1), (16,431,3,1), (17,2,1,2), (17,5,2,3),\\ & (17,135,2,1), (17,235,12,5), (17,458,3,1), (17,838,11,3), (18,143,2,1), (18,485,3,1),\\ & (19,1,3,8), (19,8,3,4), (19,45,4,3), (19,64,3,2), (19,151,2,1), (19,512,3,1), (20,1,7,19),\\ & (20,159,2,1), (20,539,3,1), (21,41,5,4), (21,167,2,1), (21,328,5,2), (21,517,32,11),\\ & (21,566,3,1), (22,175,2,1), (22,593,3,1), (23,183,2,1), (23,620,3,1), (23,986,7,2),\\ & (24,191,2,1), (24,647,3,1), (25,3,1,2), (25,199,2,1), (25,422,59,23), (25,674,3,1),\\ & (26,207,2,1), (26,493,8,3), (26,701,3,1), (27,17,6,7), (27,91,3,2), (27,215,2,1),\\ & (27,728,3,1), (28,1,1,3), (28,223,2,1), (28,755,3,1), (29,231,2,1), (29,453,5,2),\\ & (29,782,3,1), (30,239,2,1), (30,809,3,1), (31,247,2,1), (31,836,3,1), (32,255,2,1),\\ & (32,863,3,1), (33,4,1,2), (33,263,2,1), (33,580,13,5), (33,890,3,1), (34,271,2,1),\\ & (34,487,17,7), (34,793,20,7), (34,917,3,1), (35,6,5,9), (35,118,3,2), (35,162,5,3),\\ & (35,279,2,1), (35,944,3,1), (36,287,2,1), (36,971,3,1), (37,295,2,1), (37,470,7,3),\\ & (37,578,5,2), (37,998,3,1), (38,303,2,1), (38,467,30,13), (39,209,7,4), (39,311,2,1),\end{aligned}$$

$(40,319,2,1)$, $(41,5,1,2)$, $(41,244,357,197)$, $(41,327,2,1)$, $(41,467,9,4)$, $(42,335,2,1)$, $(43,1,2,7)$, $(43,145,3,2)$, $(43,343,2,1)$, $(44,13,2,3)$, $(44,351,2,1)$, $(45,359,2,1)$, $(45,703,5,2)$, $(46,109,4,3)$, $(46,367,2,1)$, $(47,3,2,5)$, $(47,375,2,1)$, $(48,383,2,1)$, $(49,6,1,2)$, $(49,391,2,1)$, $(50,399,2,1)$, $(51,20,71,97)$, $(51,172,3,2)$, $(51,407,2,1)$, $(51,823,187,74)$, $(52,415,2,1)$, $(53,423,2,1)$, $(53,828,5,2)$, $(54,431,2,1)$, $(55,2,1,3)$, $(55,439,2,1)$, $(56,447,2,1)$, $(57,7,1,2)$, $(57,455,2,1)$, $(58,463,2,1)$, $(59,199,3,2)$, $(59,471,2,1)$, $(60,479,2,1)$, $(61,487,2,1)$, $(61,953,5,2)$, $(62,287,5,3)$, $(62,495,2,1)$, $(63,503,2,1)$, $(64,511,2,1)$, $(64,813,7,3)$, $(65,1,1,4)$, $(65,8,1,2)$, $(65,519,2,1)$, $(66,527,2,1)$, $(67,226,3,2)$, $(67,535,2,1)$, $(68,543,2,1)$, $(69,551,2,1)$, $(70,559,2,1)$, $(71,21,2,3)$, $(71,28,11,15)$, $(71,567,2,1)$, $(71,756,11,5)$, $(72,575,2,1)$, $(73,9,1,2)$, $(73,173,4,3)$, $(73,220,13,9)$, $(73,299,8,5)$, $(73,583,2,1)$, $(74,591,2,1)$, $(75,253,3,2)$, $(75,599,2,1)$, $(76,607,2,1)$, $(77,615,2,1)$, $(78,623,2,1)$, $(79,631,2,1)$, $(80,639,2,1)$, $(81,10,1,2)$, $(81,647,2,1)$, $(82,3,1,3)$, $(82,225,7,5)$, $(82,655,2,1)$, $(83,35,3,4)$, $(83,280,3,2)$, $(83,663,2,1)$, $(84,43,4,5)$, $(84,671,2,1)$, $(85,166,5,4)$, $(85,679,2,1)$, $(86,687,2,1)$, $(87,695,2,1)$, $(88,19,3,5)$, $(88,703,2,1)$, $(89,11,1,2)$, $(89,335,14,9)$, $(89,412,5,3)$, $(89,711,2,1)$, $(90,719,2,1)$, $(91,307,3,2)$, $(91,727,2,1)$, $(92,357,11,7)$, $(92,735,2,1)$, $(93,743,2,1)$, $(94,751,2,1)$, $(95,122,25,23)$, $(95,759,2,1)$, $(96,767,2,1)$, $(97,12,1,2)$, $(97,775,2,1)$, $(98,29,2,3)$, $(98,783,2,1)$, $(99,334,3,2)$, $(99,791,2,1)$, $(100,237,4,3)$, $(100,799,2,1)$, $(101,807,2,1)$, $(102,815,2,1)$, $(103,3,4,13)$, $(103,69,7,8)$, $(103,552,7,4)$, $(103,823,2,1)$, $(104,5,4,11)$, $(104,831,2,1)$, $(105,13,1,2)$, $(105,839,2,1)$, $(106,847,2,1)$, $(107,361,3,2)$, $(107,855,2,1)$, $(108,863,2,1)$, $(109,4,1,3)$, $(109,871,2,1)$, $(110,879,2,1)$, $(111,887,2,1)$, $(112,895,2,1)$, $(113,14,1,2)$, $(113,23,10,17)$, $(113,903,2,1)$, $(114,911,2,1)$, $(115,388,3,2)$, $(115,919,2,1)$, $(116,537,5,3)$, $(116,927,2,1)$, $(117,935,2,1)$, $(118,43,5,7)$, $(118,663,16,9)$, $(118,943,2,1)$, $(119,694,9,5)$, $(119,951,2,1)$, $(120,959,2,1)$, $(121,15,1,2)$, $(121,967,2,1)$, $(122,975,2,1)$, $(123,415,3,2)$, $(123,983,2,1)$, $(124,991,2,1)$, $(125,37,2,3)$, $(125,999,2,1)$, $(126,1,1,5)$, $(127,301,4,3)$, $(129,2,1,4)$, $(129,16,1,2)$, $(130,51,724,989)$, $(131,442,3,2)$, $(132,665,12,7)$, $(134,25,4,7)$, $(136,5,1,3)$, $(136,235,6,5)$, $(137,17,1,2)$, $(139,469,3,2)$, $(143,662,5,3)$, $(145,18,1,2)$, $(147,62,3,4)$, $(147,496,3,2)$, $(149,291,5,4)$, $(152,45,2,3)$, $(153,19,1,2)$, $(153,980,13,7)$, $(154,365,4,3)$, $(155,523,3,2)$, $(161,20,1,2)$, $(163,6,1,3)$, $(163,550,3,2)$, $(163,647,19,12)$, $(167,895,7,4)$, $(168,517,16,11)$, $(169,21,1,2)$, $(170,787,5,3)$, $(171,577,3,2)$, $(172,11,2,5)$, $(177,22,1,2)$, $(179,53,2,3)$, $(179,604,3,2)$, $(181,71,112,153)$, $(181,429,4,3)$, $(182,1,3,17)$, $(185,23,1,2)$, $(187,631,3,2)$, $(189,811,13,8)$, $(190,7,1,3)$, $(193,3,1,4)$, $(193,24,1,2)$, $(195,658,3,2)$, $(197,114,5,6)$, $(197,912,5,3)$, $(198,811,8,5)$, $(199,316,7,6)$, $(201,25,1,2)$, $(203,685,3,2)$, $(206,61,2,3)$, $(207,71,7,10)$, $(207,568,7,5)$, $(208,493,4,3)$, $(209,26,1,2)$, $(209,107,4,5)$, $(211,89,3,4)$, $(211,712,3,2)$, $(213,46,3,5)$, $(213,52,5,8)$, $(213,416,5,4)$, $(215,151,8,9)$, $(216,17,3,7)$, $(217,1,1,6)$, $(217,8,1,3)$, $(217,27,1,2)$, $(219,739,3,2)$, $(225,28,1,2)$, $(227,766,3,2)$, $(228,37,6,11)$, $(233,29,1,2)$, $(233,69,2,3)$, $(235,557,4,3)$, $(235,793,3,2)$, $(239,653,418,299)$, $(241,30,1,2)$, $(243,820,3,2)$, $(244,9,1,3)$, $(249,31,1,2)$, $(251,2,1,5)$, $(251,847,3,2)$, $(253,21,106,243)$, $(253,567,106,81)$, $(257,4,1,4)$, $(257,32,1,2)$, $(259,874,3,2)$, $(260,77,2,3)$, $(261,451,6,5)$, $(262,23,4,9)$, $(262,621,4,3)$, $(262,631,63,47)$, $(265,13,41,112)$, $(265,33,1,2)$, $(265,104,41,56)$, $(265,832,41,28)$, $(267,901,3,2)$, $(271,10,1,3)$, $(272,793,10,7)$, $(273,34,1,2)$, $(274,409,8,7)$, $(275,116,3,4)$, $(275,928,3,2)$, $(277,541,5,4)$, $(278,323,41,39)$, $(281,35,1,2)$, $(283,955,3,2)$, $(287,85,2,3)$, $(287,118,29,39)$, $(287,524,11,9)$, $(289,36,1,2)$, $(289,685,4,3)$, $(291,982,3,2)$, $(297,19,2,5)$, $(297,37,1,2)$, $(298,11,1,3)$, $(304,467,15,13)$, $(305,38,1,2)$, $(309,713,37,28)$,

$(313, 39, 1, 2)$, $(314, 93, 2, 3)$, $(316, 749, 4, 3)$, $(321, 5, 1, 4)$, $(321, 40, 1, 2)$, $(323, 49, 8, 15)$, $(325, 12, 1, 3)$, $(329, 41, 1, 2)$, $(329, 244, 105, 116)$, $(332, 911, 7, 5)$, $(334, 171, 4, 5)$, $(337, 42, 1, 2)$, $(338, 73, 3, 5)$, $(339, 143, 3, 4)$, $(341, 101, 2, 3)$, $(341, 666, 5, 4)$, $(343, 813, 4, 3)$, $(344, 1, 1, 7)$, $(345, 43, 1, 2)$, $(351, 746, 9, 7)$, $(352, 13, 1, 3)$, $(353, 44, 1, 2)$, $(354, 19, 192, 509)$, $(356, 35, 6, 13)$, $(361, 45, 1, 2)$, $(361, 514, 9, 8)$, $(368, 109, 2, 3)$, $(369, 46, 1, 2)$, $(369, 269, 9, 10)$, $(370, 233, 6, 7)$, $(370, 877, 4, 3)$, $(376, 3, 1, 5)$, $(377, 25, 17, 42)$, $(377, 47, 1, 2)$, $(377, 200, 17, 21)$, $(377, 675, 17, 14)$, $(379, 14, 1, 3)$, $(385, 6, 1, 4)$, $(385, 48, 1, 2)$, $(386, 9, 2, 7)$, $(386, 667, 6, 5)$, $(388, 19, 15, 41)$, $(393, 49, 1, 2)$, $(395, 117, 2, 3)$, $(397, 941, 4, 3)$, $(401, 50, 1, 2)$, $(403, 170, 3, 4)$, $(405, 791, 5, 4)$, $(406, 15, 1, 3)$, $(409, 51, 1, 2)$, $(413, 239, 5, 6)$, $(415, 18, 13, 37)$, $(415, 659, 7, 6)$, $(417, 52, 1, 2)$, $(422, 1, 2, 15)$, $(422, 27, 2, 5)$, $(422, 125, 2, 3)$, $(425, 53, 1, 2)$, $(427, 174, 43, 58)$, $(433, 2, 1, 6)$, $(433, 16, 1, 3)$, $(433, 54, 1, 2)$, $(435, 187, 40, 53)$, $(435, 211, 11, 14)$, $(441, 55, 1, 2)$, $(449, 7, 1, 4)$, $(449, 56, 1, 2)$, $(449, 133, 2, 3)$, $(451, 47, 8, 17)$, $(457, 57, 1, 2)$, $(459, 235, 4, 5)$, $(460, 17, 1, 3)$, $(460, 631, 10, 9)$, $(461, 21, 5, 14)$, $(461, 168, 5, 7)$, $(463, 100, 3, 5)$, $(465, 58, 1, 2)$, $(467, 197, 3, 4)$, $(469, 916, 5, 4)$, $(473, 59, 1, 2)$, $(476, 141, 2, 3)$, $(477, 89, 4, 7)$, $(481, 60, 1, 2)$, $(487, 18, 1, 3)$, $(489, 61, 1, 2)$, $(493, 10, 3, 11)$, $(497, 62, 1, 2)$, $(501, 4, 1, 5)$, $(503, 149, 2, 3)$, $(503, 638, 105, 97)$, $(503, 765, 23, 20)$, $(505, 63, 1, 2)$, $(511, 883, 6, 5)$, $(513, 1, 1, 8)$, $(513, 8, 1, 4)$, $(513, 64, 1, 2)$, $(514, 19, 1, 3)$, $(514, 73, 12, 23)$, $(515, 39, 11, 26)$, $(515, 312, 11, 13)$, $(521, 65, 1, 2)$, $(529, 66, 1, 2)$, $(530, 157, 2, 3)$, $(531, 28, 3, 8)$, $(531, 224, 3, 4)$, $(537, 67, 1, 2)$, $(537, 788, 25, 22)$, $(541, 20, 1, 3)$, $(545, 68, 1, 2)$, $(547, 35, 2, 5)$, $(553, 69, 1, 2)$, $(557, 165, 2, 3)$, $(557, 620, 57, 55)$, $(559, 44, 3, 7)$, $(561, 70, 1, 2)$, $(568, 21, 1, 3)$, $(569, 71, 1, 2)$, $(571, 95, 11, 20)$, $(571, 429, 10, 11)$, $(571, 760, 11, 10)$, $(575, 2, 5, 33)$, $(575, 54, 5, 11)$, $(577, 9, 1, 4)$, $(577, 72, 1, 2)$, $(584, 173, 2, 3)$, $(584, 299, 4, 5)$, $(585, 73, 1, 2)$, $(588, 127, 3, 5)$, $(593, 13, 89, 318)$, $(593, 74, 1, 2)$, $(593, 104, 89, 159)$, $(593, 351, 89, 106)$, $(595, 22, 1, 3)$, $(595, 251, 3, 4)$, $(601, 75, 1, 2)$, $(601, 933, 22, 19)$, $(609, 76, 1, 2)$, $(611, 181, 2, 3)$, $(615, 412, 7, 8)$, $(617, 77, 1, 2)$, $(622, 23, 1, 3)$, $(625, 78, 1, 2)$, $(626, 5, 1, 5)$, $(629, 364, 5, 6)$, $(633, 79, 1, 2)$, $(638, 7, 2, 9)$, $(638, 189, 2, 3)$, $(641, 10, 1, 4)$, $(641, 80, 1, 2)$, $(642, 71, 12, 25)$, $(649, 3, 1, 6)$, $(649, 24, 1, 3)$, $(649, 81, 1, 2)$, $(651, 1, 3, 26)$, $(651, 8, 3, 13)$, $(657, 82, 1, 2)$, $(659, 278, 3, 4)$, $(665, 83, 1, 2)$, $(665, 197, 2, 3)$, $(672, 43, 2, 5)$, $(673, 84, 1, 2)$, $(676, 25, 1, 3)$, $(681, 85, 1, 2)$, $(687, 2, 1, 7)$, $(689, 86, 1, 2)$, $(692, 205, 2, 3)$, $(697, 87, 1, 2)$, $(701, 47, 13, 32)$, $(701, 376, 13, 16)$, $(703, 26, 1, 3)$, $(705, 11, 1, 4)$, $(705, 88, 1, 2)$, $(709, 363, 4, 5)$, $(712, 335, 7, 9)$, $(713, 89, 1, 2)$, $(713, 154, 3, 5)$, $(713, 449, 6, 7)$, $(719, 213, 2, 3)$, $(721, 90, 1, 2)$, $(723, 305, 3, 4)$, $(725, 177, 5, 8)$, $(729, 17, 2, 7)$, $(729, 91, 1, 2)$, $(730, 1, 1, 9)$, $(730, 27, 1, 3)$, $(737, 92, 1, 2)$, $(745, 93, 1, 2)$, $(746, 221, 2, 3)$, $(751, 6, 1, 5)$, $(753, 94, 1, 2)$, $(757, 28, 1, 3)$, $(761, 95, 1, 2)$, $(764, 131, 5, 9)$, $(769, 12, 1, 4)$, $(769, 96, 1, 2)$, $(773, 229, 2, 3)$, $(777, 97, 1, 2)$, $(784, 29, 1, 3)$, $(785, 98, 1, 2)$, $(786, 563, 17, 19)$, $(787, 332, 3, 4)$, $(793, 99, 1, 2)$, $(797, 51, 2, 5)$, $(800, 237, 2, 3)$, $(801, 100, 1, 2)$, $(804, 293, 5, 7)$, $(806, 77, 16, 35)$, $(807, 442, 9, 11)$, $(809, 101, 1, 2)$, $(811, 30, 1, 3)$, $(813, 173, 40, 67)$, $(817, 102, 1, 2)$, $(820, 153, 4, 7)$, $(824, 3, 2, 13)$, $(825, 103, 1, 2)$, $(827, 245, 2, 3)$, $(832, 5, 2, 11)$, $(833, 13, 1, 4)$, $(833, 104, 1, 2)$, $(834, 427, 4, 5)$, $(838, 31, 1, 3)$, $(838, 181, 3, 5)$, $(841, 105, 1, 2)$, $(845, 489, 5, 6)$, $(849, 106, 1, 2)$, $(851, 359, 3, 4)$, $(854, 253, 2, 3)$, $(857, 107, 1, 2)$, $(865, 4, 1, 6)$, $(865, 32, 1, 3)$, $(865, 108, 1, 2)$, $(873, 109, 1, 2)$, $(876, 7, 1, 5)$, $(881, 110, 1, 2)$, $(881, 261, 2, 3)$, $(889, 111, 1, 2)$, $(892, 33, 1, 3)$, $(897, 14, 1, 4)$, $(897, 112, 1, 2)$, $(902, 71, 3, 7)$, $(904, 23, 5, 17)$, $(905, 113, 1, 2)$, $(908, 269, 2, 3)$, $(913, 114, 1, 2)$, $(915, 386, 3, 4)$, $(919, 34, 1, 3)$, $(921, 115, 1, 2)$, $(922, 59, 2, 5)$, $(929, 116, 1, 2)$, $(935, 277, 2, 3)$, $(937, 117, 1, 2)$, $(945, 118, 1, 2)$, $(946, 35, 1, 3)$, $(953, 119, 1, 2)$, $(959, 491, 4, 5)$, $(961, 15, 1, 4)$, $(961, 120, 1, 2)$, $(962, 285, 2, 3)$, $(963, 26, 3, 10)$, $(963, 208, 3, 5)$, $(969, 121, 1, 2)$, $(973, 36, 1, 3)$, $(974, 251, 7, 11)$, $(977, 75, 17, 40)$,

$(977, 122, 1, 2)$, $(977, 600, 17, 20)$, $(979, 413, 3, 4)$, $(985, 123, 1, 2)$, $(989, 293, 2, 3)$, $(989, 388, 153, 209)$, $(991, 87, 4, 9)$, $(993, 124, 1, 2)$, $(1000, 37, 1, 3)$,

Proof. Note that (x, y) is a solution to $Ax^3 - By^3 = 1$ with $x, y \leq 0$, if and only if $(|y|, |x|)$ is a solution to $Bx^3 - Ay^3 = 1$. Hence it suffices to find all solutions (x, y) of $Ax^3 - By^3 = 1$ with $A < B$.

We then run the following MAGMA script to find all such solutions:

```
I := Integers ();
k<x> := PolynomialRing(I);

solutions := <>;

for A in [1..1000] do
  for B in [A..1000] do

    if Gcd (A,B) ne 1 or A eq B+1 or A eq B-1 then
      continue;
    end if;

    f := A * x^3 - B;
    T := Thue (f);

    s := Solutions (T,1);

    if s ne [] and s[1][1] ne 0 and s[1][2] ne 0 then
      Append (~solutions, <A,B,s>);
    elif #s eq 2 and s[2][1] ne 0 and s[2][2] ne 0 then
      Append (~solutions, <A,B,s>);
    end if;

  end for;
end for;
```

$\square$

** This list is incomplete as my program is still running. However, the program loops from A=1 to 1000, and B=A to 1000 and is currently at A=601. Since B>A, we have thus classified all solutions when A,B < 601, which is enough for our proof of k=14. I will keep updating this.

For the cases when $d_1 \neq 1$, (6.1) shows that $\alpha_1 w_1 - \alpha_0 w_0 = 1$ has a solution only when $D_1 = 1$ and $\alpha_1 = 3, \alpha_0 = 10$ or when $D_1 = 7$ and $\alpha_1 = 1, \alpha_0 = 6$ and $D_1 \mid x + 1$. In the former, we have that $x = \alpha_0 u_0^3 = 80$ and

$$S_{14}(80) = 2^4 \cdot 3^5 \cdot 5^2 \cdot 7^2 \cdot 23 \cdot 421 \cdot 3359 \cdot 149297 \cdot 10008516307$$

which isn't a third power. In the latter, we have $x = \alpha_0 u_0^3 = 48$ and

$$S_{14}(48) = 2^4 \cdot 3^2 \cdot 5 \cdot 7^3 \cdot 97 \cdot 109 \cdot 353 \cdot 751 \cdot 166886978459$$

which isn't a third power.

Finally, Jacobson, Pintér, and Walsh [JPW03] have verified Schäffer's conjecture when $k \le 58$ and $n = 2$. Thus, since we have shown that there are no solutions to the system (14), (15), (16), by Proposition 2.4 there are no solutions to $S_k(x) = y^n$ for $k = 4, 6, 8, 14$.

7. Conclusions

We have shown the cases $2 \le k \le 14$ even, except $k = 12$, of Schäffer's conjecture by systematically applying a more detailed descent, the modular method plus local methods, and then using prior results on ternary equations with coefficients.

The case $k = 12$ remains outside the reach of this approach as the levels in the modular method become too big.

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